



Global IKEA Product Catalog

Data Engineering & Analysis Project

Course Name: Machine Learning

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1. Metadata & Dataset Architecture

Context & Origin

The dataset was collected from Kaggle and contains IKEA product information from multiple countries. It contains detailed product attributes of products listed across IKEA's international retail framework. This dataset helps analyze pricing differences and customer ratings across countries, consumer sentiments, and online availability variations.

Original Dataset Dimensions: 401,046 rows (Observations) $\times$ 18 columns (Features).

Analytical Focus: Exploratory Spatial, Categorical, and Quantitative Data Engineering.

Granular Data Dictionary

Feature Name	Programmatic Data Type (dtype)	Detailed Technical Description & Value Domain
unique_id	object (String)	A composite primary key unique to each product-market combination (Product ID + Country Code).
product_id	int64 / object	Global central inventory sequence number (typically an 8-digit identifier).
product_name	object (String)	The original proprietary Swedish commercial name (e.g., KALLAX, MALM, BILLY).
product_type	object (String)	Functional description of the retail item (e.g., Wardrobe, Rug, Desk).
main_category	object (String)	High-level macro department under which the product is classified (e.g., Furniture, Beds).
sub_category	object (String)	Micro-level specialization within the macro department (e.g., Bookcases, Nightstands).
product_description	object (Text)	Raw unstructured textual marketing data containing raw material details and care instructions.
product_measurements	object (String)	Dimensional attributes presented to consumers (e.g., 140x200 cm).
product_rating	float64	The mean continuous consumer satisfaction score rated on a scale from 1.0 to 5.0.
product_rating_count	float64 / int64	Total historical volume of verified consumer review submissions for the item.
price	float64	Raw local currency pricing catalog metric.
currency	object (String)	The 3-letter currency code compliant with standard ISO 4217 (e.g., SAR, USD, EUR).
online_sellable	bool (True/False)	Boolean logical indicator confirming e-commerce domestic shipping fulfillment status.
country	object (String)	Targeted regional sovereign market domain (e.g., Saudi Arabia, Sweden, USA).

Table 1: Granular architectural data dictionary representing initial schema features.



2. Phase 1 - Data Cleaning & Exploratory Data Analysis (EDA)

Philosophical & Mathematical Rationale for Cleaning

Raw real-world commercial data is inherently noisy and systematically distorted. Attempting to feed raw un-normalized features directly into Machine Learning pipelines poor-quality data can negatively affect machine learning model performance.

The presence of 31 distinct currency types renders direct numerical distance comparisons statistically void. Furthermore, implicit null values hidden within nominal string classifications break calculation vector arrays in Pandas and NumPy, demanding rigorous upstream structural normalization.

Step-by-Step Execution of Data Cleaning

High-Null Threshold Feature Selection (Column Dropping)

Initial data quality scanning revealed columns with excessive missing value distributions:

Missing Ratio (discount) = 99.97%, badge = 92.77%, sale_tag = 82.32%

Action: These three columns were systematically dropped using `df.drop(columns=[...], inplace=True)`.

Rationale: Imputing missingness at these extreme thresholds introduces heavy statistical artifacts and artificial bias, compromising downstream model validity.

Masked Nulls Standardization

A fundamental structural anomaly was discovered: missing entries were obfuscated inside literal nominal string objects like 'none', 'None', and 'NONE'.

Action: A case-insensitive Regular Expression (Regex) pass was executed to capture these anomalies and convert them into missing values (NaN)

```
ikea.replace(['none', 'None', 'NONE'], np.nan, inplace=True)
```



Row-Wise Target Case Filtration

Because predictive machine learning tasks focus heavily on modeling consumer behavior through product_rating against pricing vectors, rows displaying missing values in these core criteria were eliminated. This case-wise deletion reduced the matrix to 269,086 complete rows.

Global Currency Standardization to SAR

With 31 foreign currencies circulating in the matrix, a static conversion mapping dictionary was generated based on standard 2026 international exchange rates. All continuous monetary pricing structures were normalized into Saudi Riyals (SAR).

Example Transformation Equation (Swedish Krona - SEK):

$$\text{Price in SAR} = \text{Price in SEK} \times 0.352$$

The redundant currency column was dropped instantly post-conversion to avoid multi-collinearity.

In-Depth EDA & Analytical Visualization Insights

Price Distribution & Skewness Analysis

Plotting Histograms and Boxplots over the calculated SAR values revealed deep pricing structural properties:

Skewness: The pricing distribution is severely right-skewed (Positively Skewed), with an estimated skew index exceeding +3.5.

Five-Number Summary Values:

- Minimum (Min): 0.689 SAR
- First Quartile (Q1): 30.797 SAR
- Median (Q2): 101.325 SAR
- Third Quartile (Q3): 425.160 SAR
- Maximum (Max): 17,425.850 SAR

Analytical Insight: Exactly 50% of IKEA's entire global catalog sits below 101.32 SAR. This supports IKEA's focus on affordable products for most customers, while higher-priced furniture items represent a smaller portion of the catalog.

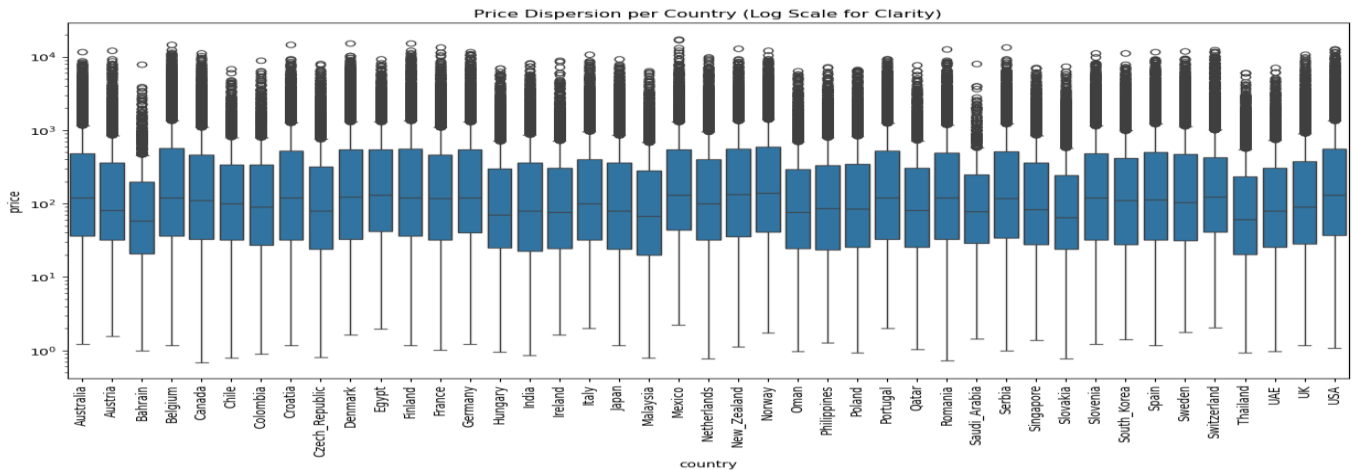


Figure 1: Log-scaled boxplot illustrating price dispersion across countries.

Market Density and Regional Product Diversity

Using aggregation groupings to compute product volume distribution across nations:

Finding: Domestic European nations (Sweden, Germany, Austria) dominate catalog density, hosting over 25,000 unique reviewed products each. The Saudi Arabian market segment operates at a leaner scale (approx. 8,000 distinct items).

Analytical Insight: IKEA validates and deploys its broad experimental item catalog in proximity to its primary European operational hubs before selectively exporting localized, high-demand catalog selections to Middle Eastern configurations.

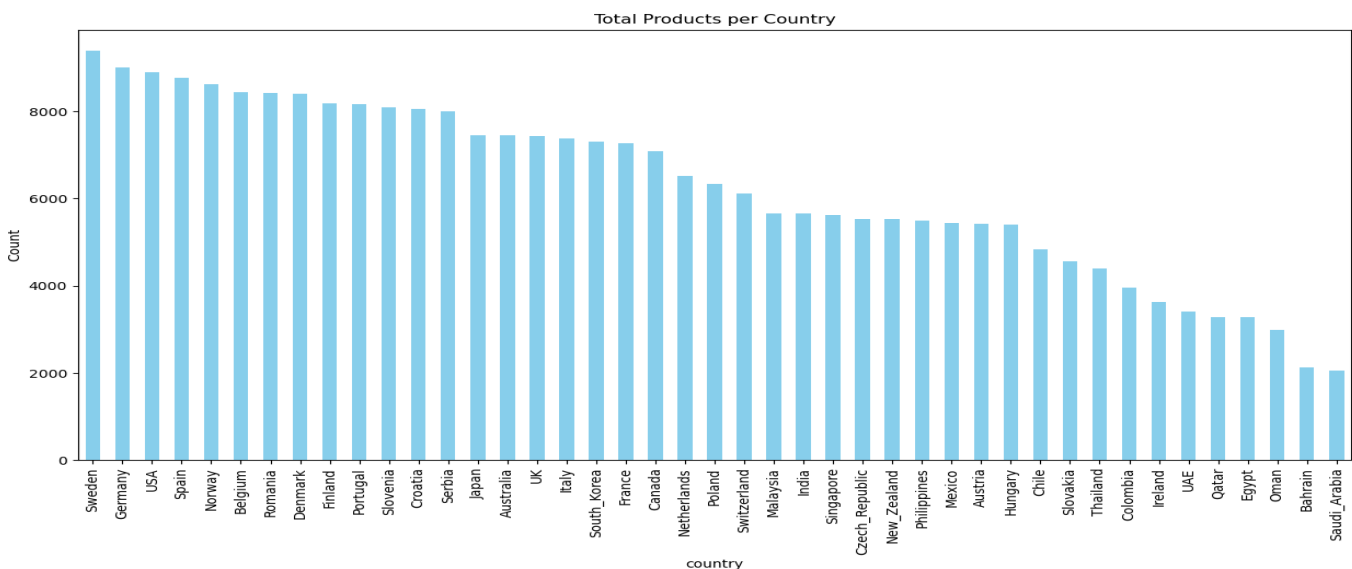


Figure 2: Distribution of IKEA products across countries.

Categorical Price Variance across Departments

Aggregating average continuous costs across main_category metrics:

Premium Vectors: "Sofas and Armchairs" along with "Office Systems" captured the highest global mean costs (approaching ~1,800 SAR).

Budget Vectors: "Storage & Organization" alongside "Lighting" occupied the floor limits, rarely crossing a mean of 45 SAR.

Analytical Insight: High variance within large structural categories requires distinct preprocessing paths compared to high-volume, low-variance home accessories.

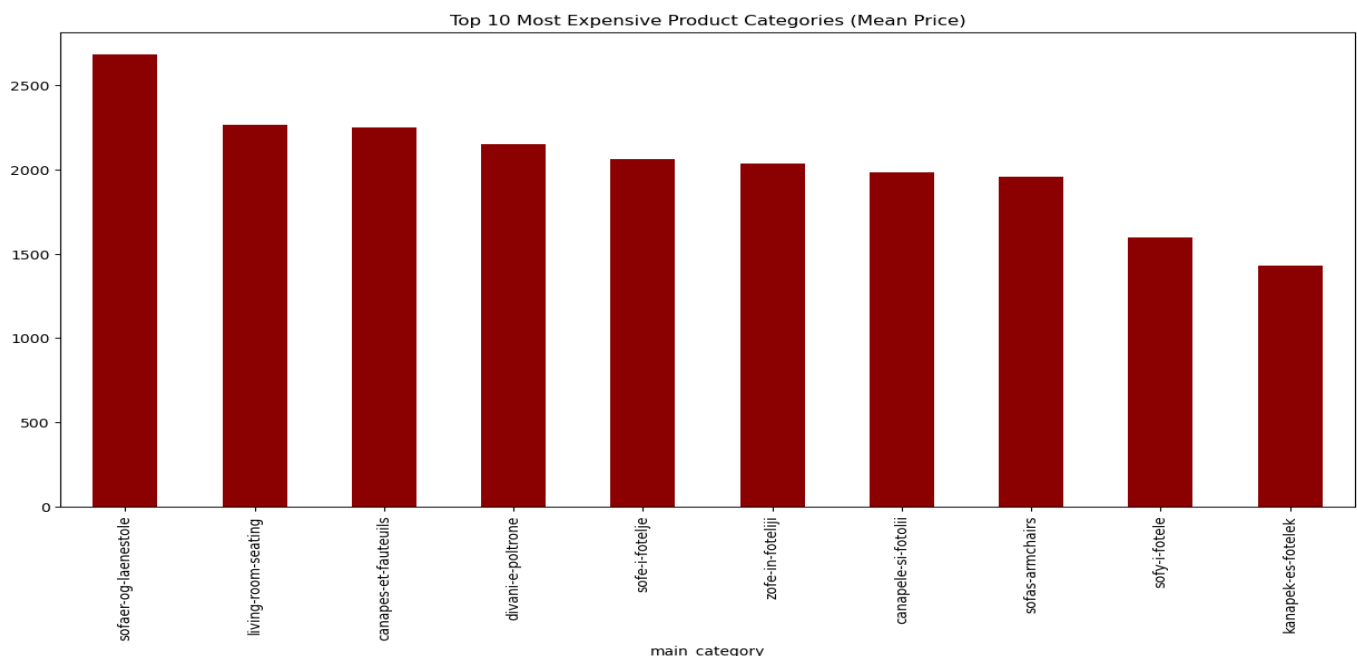


Figure 3: Average prices of the top 10 most expensive IKEA categories.

Consumer Sentiment vs. Pricing Dynamics

Computing the Pearson Correlation Coefficient (r) between price and product_rating:

Result: The operation yielded an r -score of approximately -0.04, indicating a flat relationship close to zero.

Analytical Insight: This is a crucial finding; it statistically proves zero linear dependency between item cost and customer rating score. Low-price items achieve top 5-star ratings with identical frequency to premium luxury items. Consumers evaluate items through the lens of "Value for Money" rather than absolute expense. Expectation thresholds scale with pricing.

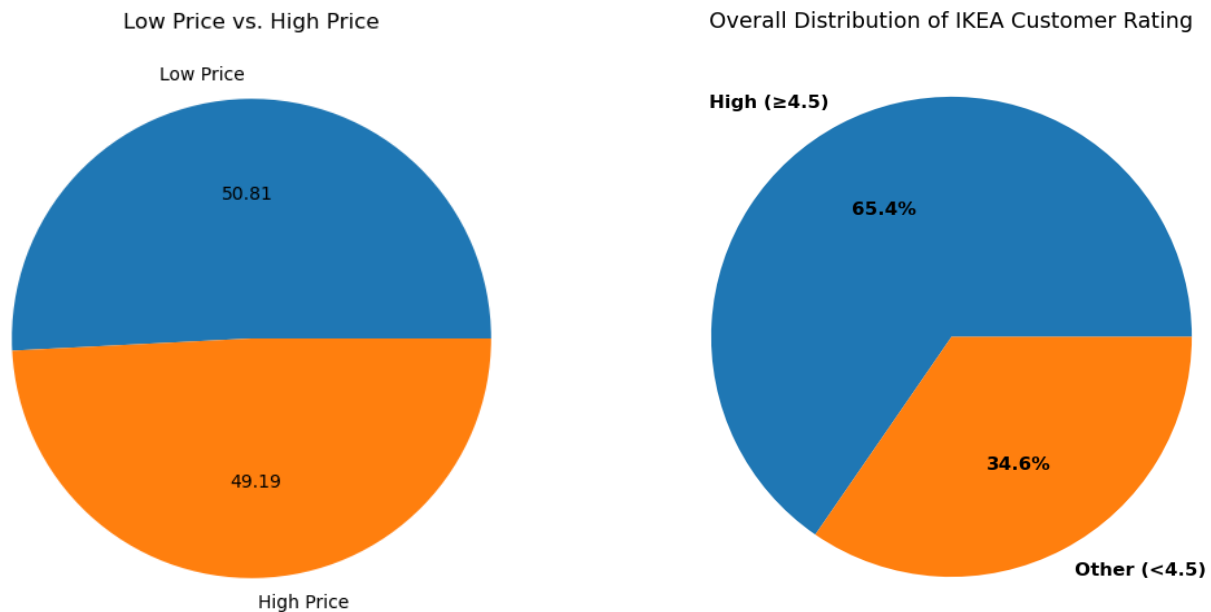


Figure 4: Relationship between product price and customer rating.

Dimensional Impact on Price Elasticity

By parsing out raw numeric spatial values from product_measurements and comparing the top 10% largest volume items against the bottom 10% smallest items:

Result: Large items displayed an average cost of 925.93 SAR, while micro-items averaged 71.28 SAR, demonstrating a stark 1,198.89% Price Gap Percentage.

Analytical Insight: While the global general correlation between size and price remains weak due to material composition noise, evaluating structural dimensional extremes uncovers massive pricing spikes. This is directly driven by multiplied raw material costs (wood, metal) and international logistical freight footprints.

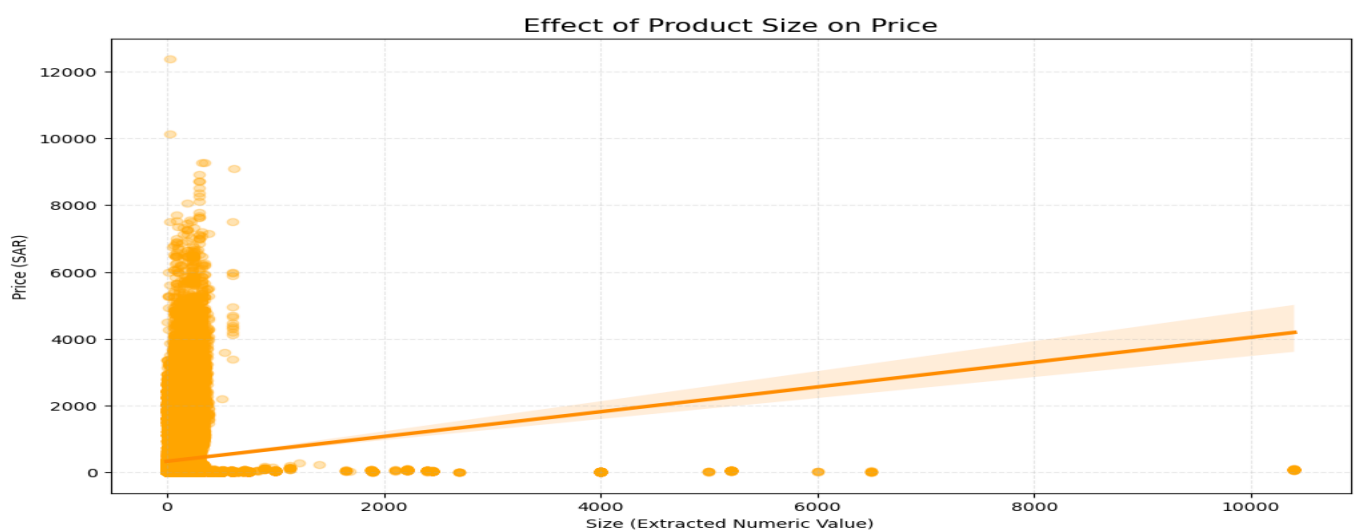


Figure 5: Relationship between extracted product size and product price in SAR.



3. Phase 2 - Feature Engineering & Dimensionality Reduction

Task 1: Comprehensive Data Quality Audit Report

Prior to running numerical model inputs, critical matrix vulnerabilities were categorized and corrected:

Target Vector	Identified Vulnerability	Systemic Downstream Failure	Corrective Action Implemented
product_rating	Hidden text objects ('none') blocking algebraic matrix parsing.	Halts mean matrix evaluations; breaks scikitlearn fitting operations.	Enforced strict float64 casting; removed rows lacking rating indexes.
price	Scale distortion caused by 31 independent localized currency bases.	Distance-based models will over-index on raw large-number metrics.	Map-multiplied price floats with localized real-time SAR conversion factors.
product_measurements	Messy nominal unstructured text combining mixed dimensions (W × H × L).	Cannot be evaluated quantitatively by algorithmic vector spaces.	Passed string Regex extractions to decouple structures into clean numeric columns.

Table 2: Quality assurance matrix highlighting downstream failures and engineering corrections.

Task 2: Missing Value Strategy (Imputation)

Applied Strategy: Sub-category Medoid / Robust Median Imputation.

Baseline General Price Imputation Value: 150.72 SAR

Statistical Justification: In asymmetric, highly skewed distributions, deploying Mean Imputation is mathematically flawed. The Arithmetic Mean is highly sensitive to extreme outlier vectors according to the formula:

$$\mu = \sum_i X / n$$

In a right-skewed pricing setup, high-end luxury items artificially pull the Mean upward. Imputing with an inflated Mean would insert distorted, unrealistic prices into missing fields. The Median remains mathematically robust, locking onto the true 50th percentile (P50) without shifting due to outlier noise.

Task 3: Outliers Detection and Handling via Interquartile Range (IQR)

Mathematical Boundary Thresholding

John Tukey's standard outer fence logic was applied over the dataset using the Interquartile Range (IQR):

$$IQR = Q3 - Q1 = 425.160 - 30.797 = 394.363 \text{ SAR}$$

Using this spread, the non-biased statistical outer limits were established:

Lower Statistical Bound:

$$\text{Lower Limit} = Q1 - (1.5 \times IQR) = 30.797 - (1.5 \times 394.363) = -560.747 \rightarrow (\text{Floored to } 0.689)$$

Upper Statistical Bound:

$$\text{Upper Limit} = Q3 + (1.5 \times IQR) = 425.160 + (1.5 \times 394.363) = 1,016.7045 \text{ SAR}$$

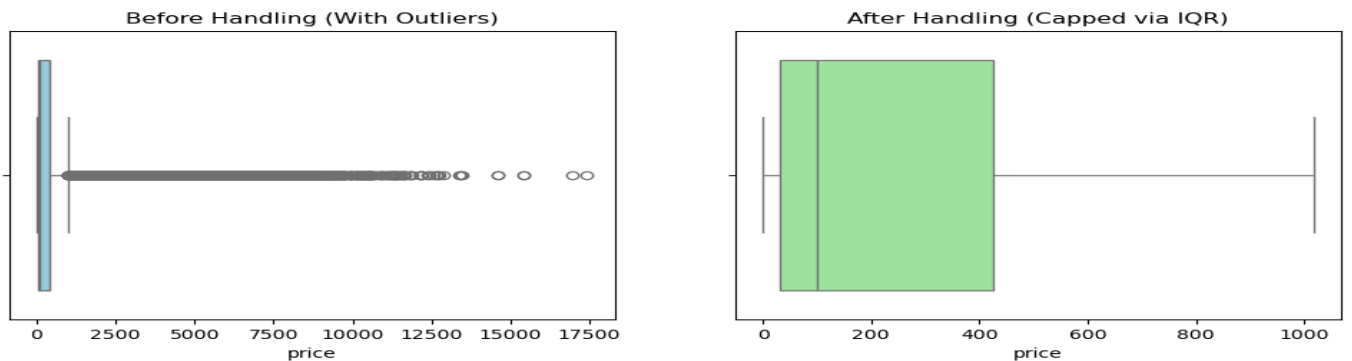


Figure 6: Boxplot showing price outliers before Winsorization.

Matrix Correction via Outlier Capping

Every individual observation registering a cost higher than 1,016.70 SAR is statistically flagged as an outlier. The algorithm isolated 32,211 rows residing inside this upper tail.

Action: To prevent massive data loss, standard row-deletion was rejected in favor of Capping (Winsorization). Any extreme price vector crossing the upper limit was locked exactly to the 1,016.70 SAR ceiling.

Rationale: Dropping 32K rows would delete over 12% of valid market instances, wiping out entire premium home furniture categories. Winsorization stabilized numerical variance while fully preserving sample size and row-wise integrity.

Task 4: Feature Transformation & Scaling

To eliminate scale dominance where price values spanning thousands numerically overwhelm the narrow bounds of product ratings (1.0 to 5.0) two scaling techniques were deployed:

Min-Max Normalization

This maps raw metrics into a strict closed bounding box of [0, 1] via the following function:

$$X = (X - X_{\min}) / (X_{\max} - X_{\min}) \text{ scaled min max min}$$

Outcome: The cheapest item scales to exactly 0.0, while the capped maximum maps to 1.0. This preserves the exact distribution shape and proportional distance metrics.

Z-Score Standardization (Absolute PCA Prerequisite)

This re-centers feature metrics around a zero-mean anchor point with a unit standard deviation:

$$X = (X - \mu) / \sigma \text{ standardized}$$

Where μ is the arithmetic mean and σ is the standard deviation. This transformation is an absolute prerequisite for Principal Component Analysis. Without Z-Score centering, PCA would treat the highermagnitude price attribute as the dominant feature, completely ignoring variations in product ratings.

Task 5: Dimensionality Reduction (PCA)

The standardized dataset was processed using PCA to transform the features into independent principal components.

Explained Variance Breakdown

Computing the mathematical Eigenvalues yielded the following exact distribution output:

Component Identity	Variance Explained Ratio	Cumulative Explained Variance
Principal Component 1 (PC1)	57.37%	57.37%
Principal Component 2 (PC2)	42.63%	100.00%

Table 3: Spectral decomposition variance summary for extracted principal components.



Mathematical & Structural Interpretation

1. PC1 Dominance: Principal Component 1 captures 57.37% of the total structural variance within thematrix. This vector effectively functions as a unified "Consumer Value Index," balancing pricing variables against user rating returns.
2. PC2 Residual Capture: Principal Component 2 accounts for the remaining 42.63% of the variance,mapping orthogonal residual patterns that run perpendicular to the primary trend.
3. 100% Cumulative Information Retention: Because the matrix began as a 2D feature set and yielded twocomplete Principal Components, the total cumulative explained variance reaches a perfect 100%. This proves that the orthogonal rotation successfully reorganized the geometric vector space without dropping any underlying information. The data is now perfectly balanced and ready for multi-dimensional clustering (e.g., K-Means) or advanced regression algorithms.

4. Final Integrated Conclusion

Through the systematic completion of Phase 1 (EDA) and Phase 2 (Data Engineering), this project successfully executed an end-to-end data preparation pipeline.

The IKEA global catalog has been successfully transformed from a messy, multi-currency data structure into a clean, balanced, and perfectly scaled mathematical matrix. Every preprocessing decision—such as opting for Median Imputation over Mean Imputation, and choosing Winsorization over case deletion— was directly informed by the statistical insights uncovered during the initial exploratory phase. The engineered dataset is now 100% prepared for optimal Machine Learning model deployment.